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(54) **DISHWASHER**

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(57) **ABSTRACT**

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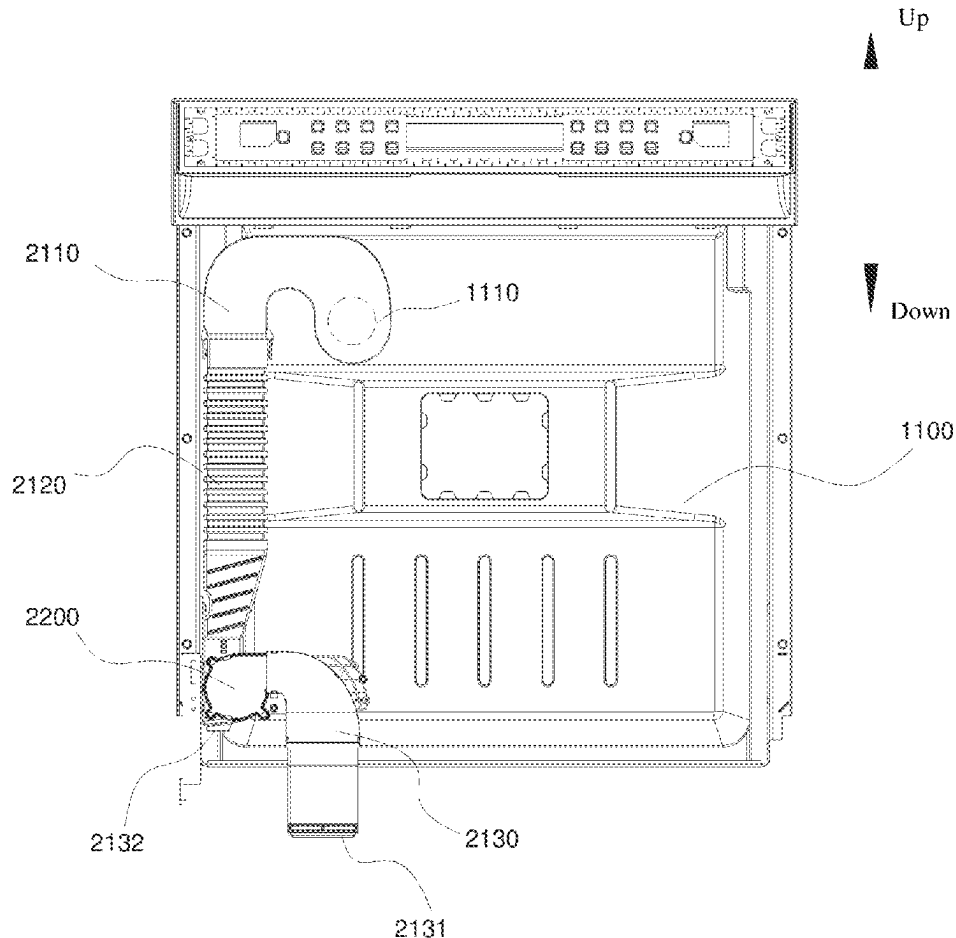
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Provided is a dishwasher. The dishwasher includes an outer door; an inner door having a first through hole; and an air discharge device disposed between the outer door and the inner door, the air discharge device including an air duct and a fan, the air duct being in communication with a washing chamber of the dishwasher through the first through hole, the air duct having a water return opening upstream of the fan, and the water return opening being configured to allow external air to enter the air duct from the water return opening during an operation of the fan and to discharge condensate water into the washing chamber.



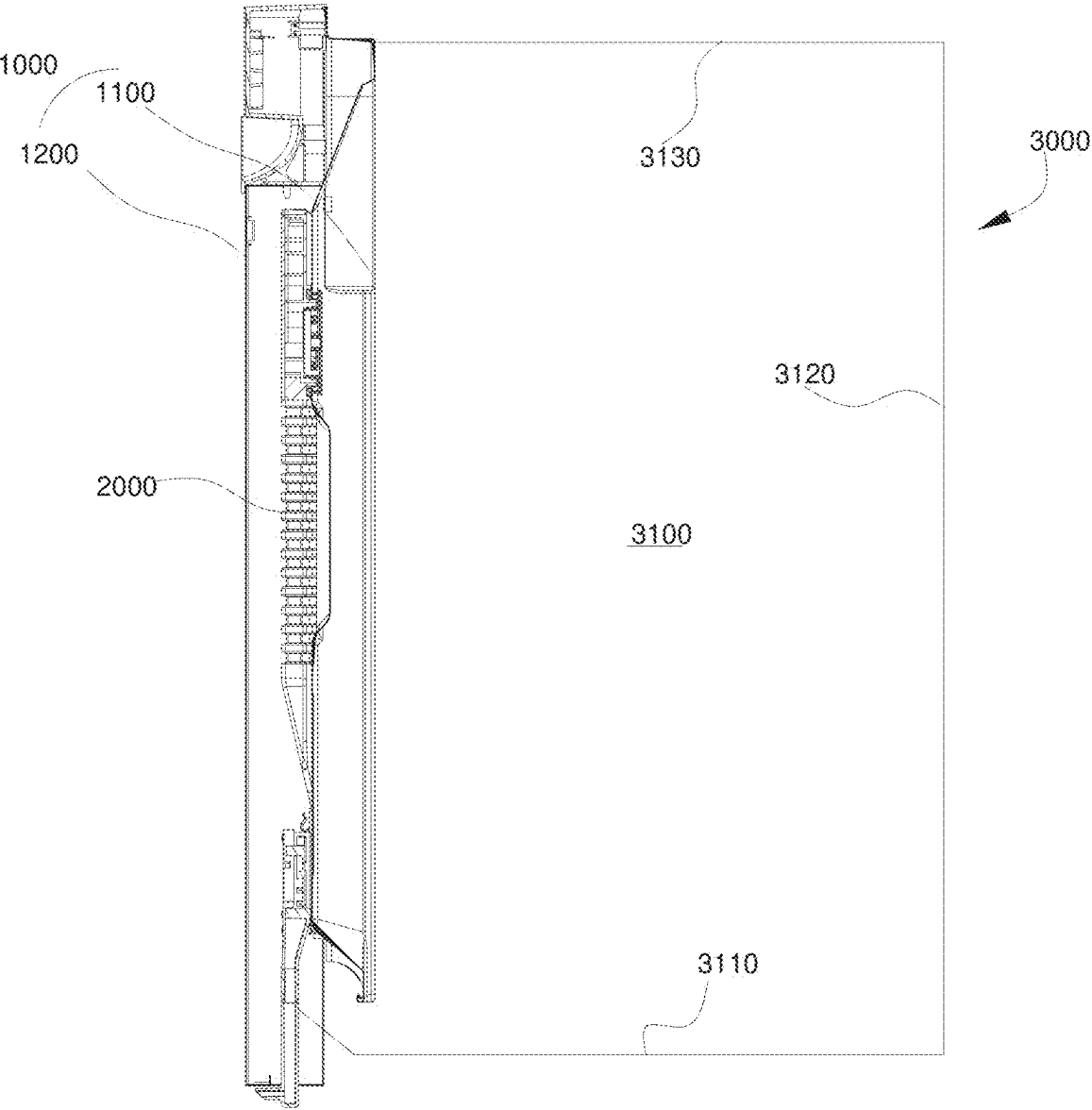


FIG. 1

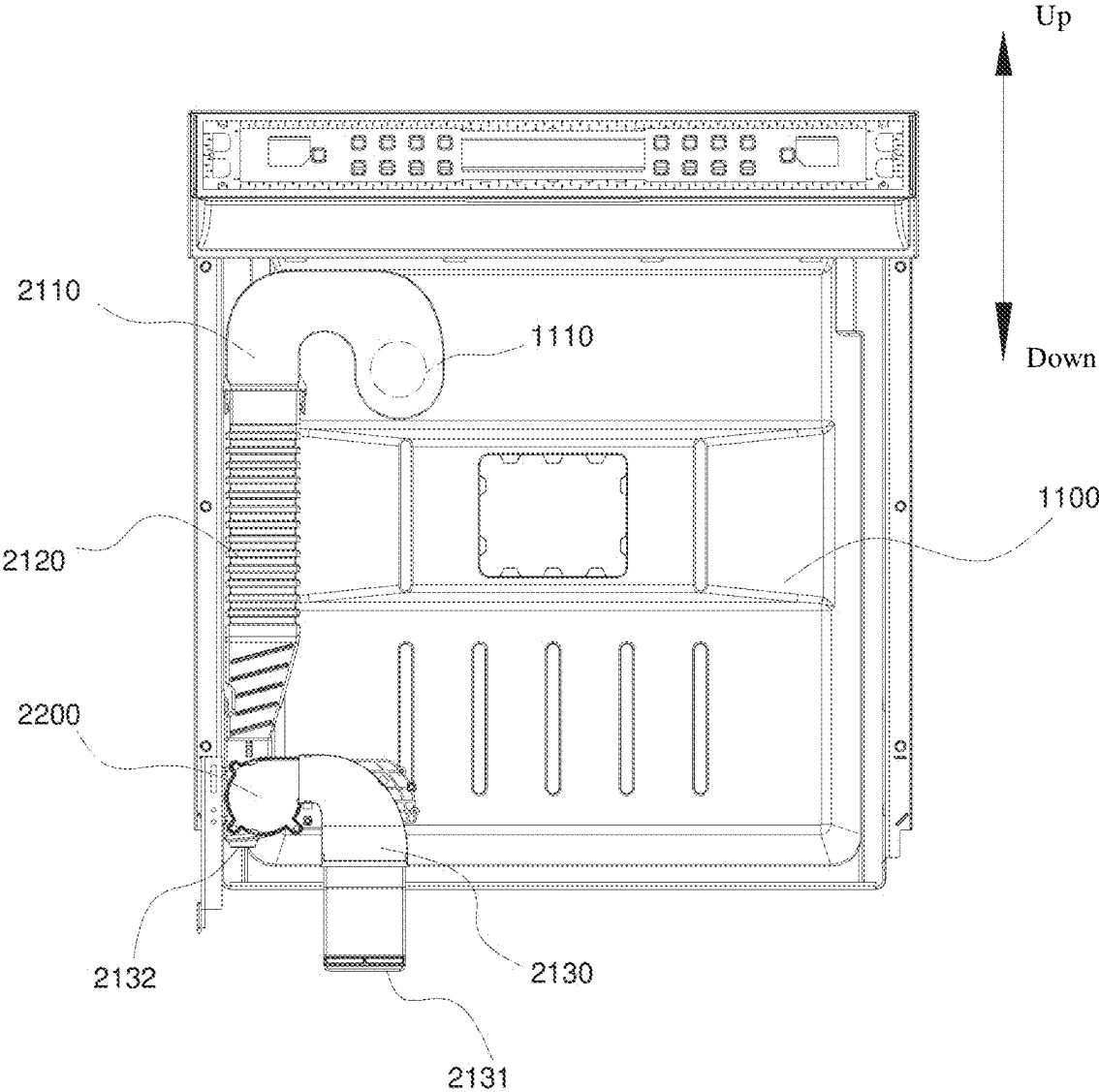


FIG. 2

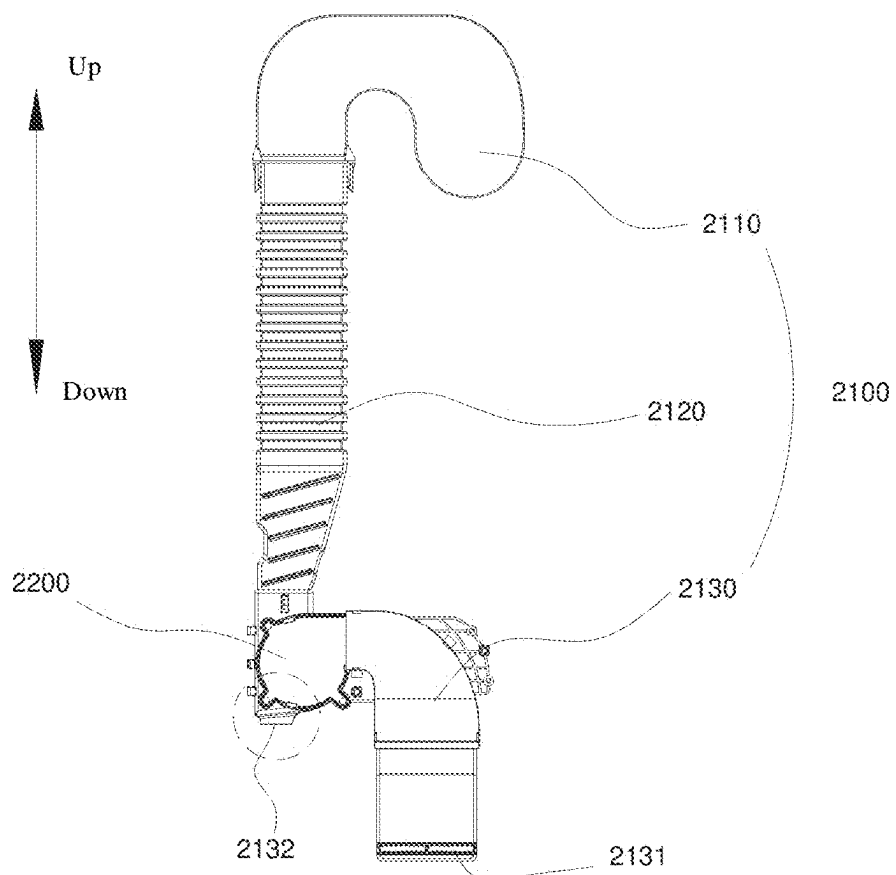


FIG. 3

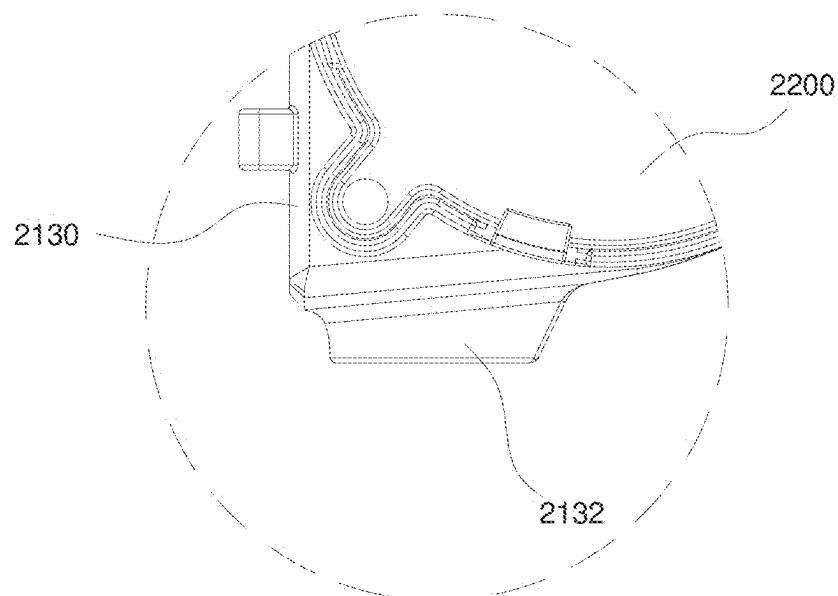


FIG. 4

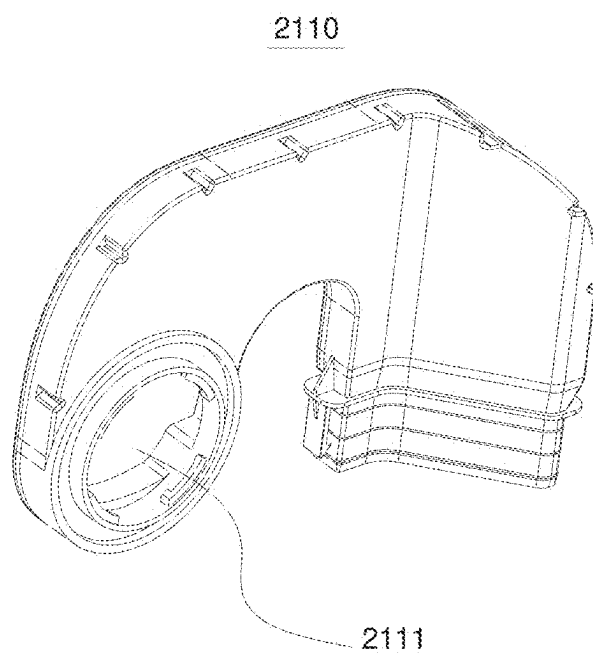


FIG. 5

2120

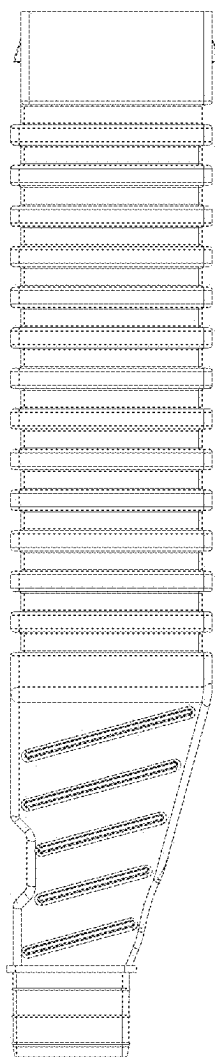


FIG. 6

DISHWASHER

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of International Application No. PCT/CN2023/097599 filed on May 31, 2023, which is based on and claims priorities to Chinese Patent Application Nos. 202211468315.X and 202223111273.2 both filed on Nov. 22, 2022, the entire contents of each of which are incorporated herein by reference. No new matter has been introduced.

FIELD

[0002] The present disclosure relates to the field of cleaning device technologies, and more particularly, to a dishwasher.

BACKGROUND

[0003] In a drying phase, a dishwasher needs to use an air discharge device to extract air from a washing chamber, thereby drying the dishes. However, in an early stage of the drying phase, air in the washing chamber is at a high temperature and high humidity. Directly extracting the air from the washing chamber with the air discharge device results in the expelled air being at a high temperature, which poses a predetermined risk, and leads to generation of a significant amount of condensate water at an air discharge opening or in an external environment.

SUMMARY

[0004] The present disclosure aims to solve technical problems in the related art to some extent. To this end, the present disclosure provides a dishwasher, which is capable of effectively reducing a temperature of the expelled air and a probability of condensate water forming at an air discharge opening and outside the dishwasher.

[0005] To achieve the above objective, the present disclosure provides a dishwasher. The dishwasher includes: an outer door; an inner door connected to the outer door and having a first through hole; and an air discharge device disposed between the outer door and the inner door, the air discharge device including an air duct and a fan, the air duct being in communication with a washing chamber of the dishwasher through the first through hole, the fan being adapted to draw air from the washing chamber into the air duct and discharge the air out of the air duct, the air duct having a water return opening upstream of the fan, and the water return opening being configured to allow external air to enter the air duct from the water return opening during an operation of the fan and to discharge condensate water into the washing chamber.

[0006] In some embodiments of the present disclosure, the dishwasher includes a tub. The tub includes a bottom plate enclosing the washing chamber. The water return opening is located above the bottom plate and spaced apart from the bottom plate.

[0007] In some embodiments of the present disclosure, the water return opening is located directly above the bottom plate.

[0008] In some embodiments of the present disclosure, the condensate water drips from the water return opening onto the bottom plate.

[0009] In some embodiments of the present disclosure, the water return opening is open downwardly.

[0010] In some embodiments of the present disclosure, the water return opening is in communication with a space between the inner door and the outer door.

[0011] In some embodiments of the present disclosure, the water return opening is located below an inlet of the fan.

[0012] In some embodiments of the present disclosure, an open direction of the water return opening spatially intersects with an air intake direction of the fan.

[0013] In some embodiments of the present disclosure, the water return opening is arranged adjacent to the fan.

[0014] In some embodiments of the present disclosure, the air duct includes a condensation segment extending in an up-down direction, the water return opening being located below the condensation segment.

[0015] In some embodiments of the present disclosure, an open direction of the water return opening is the same as an extension direction of the condensation segment.

[0016] In some embodiments of the present disclosure, the air duct includes an air intake segment, a condensation segment, and an air discharge segment that are sequentially in communication with each other, the air intake segment being constructed as an inverted U-shape and in communication with the washing chamber, the fan being disposed in the air discharge segment, and the water return opening being disposed in one of the air discharge segment and the condensation segment.

[0017] In some embodiments of the present disclosure, the air duct includes an air intake tube forming the air intake segment, a condensate tube forming the condensate segment, and an air discharge tube forming the air discharge segment, and the fan and the water return opening being located in the air discharge segment.

[0018] In some embodiments of the present disclosure, the fan is located below $\frac{1}{2}$ or $\frac{1}{3}$ of a height of the inner door, and the air discharge segment has an air discharge opening located below the inner door and the outer door.

[0019] In technical solutions of the present disclosure, with the air discharge device between the outer door and the inner door, the air in the washing chamber is effectively extracted and discharged, and thus drying of the dishes is realized. The air discharge device has the water return opening. The water return opening is configured to discharge the condensate water, and when the fan is in operation, the external air can enter the air duct through the water return opening and mix with high-temperature and high-humidity air extracted into the washing chamber in the air duct. In this way, an air discharge temperature of the air discharge device and a risk can be reduced, and moisture in the air when the air is mixed can be further removed. The formed condensate water is discharged back to the washing chamber through the water return opening, so as to reduce a probability of generation of the condensate water in an external environment. By disposing the water return opening, both recirculation of condensate water and mixing of air to lower the air discharge temperature are realized.

[0020] Additional aspects and advantages of the present disclosure will be provided at least in part in the following description, or will become apparent at least in part from the following description, or can be learned from practicing of the present disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] The above and/or additional aspects and advantages of the present disclosure will become more apparent and more understandable from the following description of embodiments taken in conjunction with the accompanying drawings.

[0022] FIG. 1 is a schematic view of mounting of an air discharge device according to some embodiments of the present disclosure.

[0023] FIG. 2 is a schematic view of mounting of an air discharge device according to some embodiments of the present disclosure.

[0024] FIG. 3 is a schematic view of an air discharge device according to some embodiments of the present disclosure.

[0025] FIG. 4 is an enlarged view of a region indicated by a dashed line in FIG. 3.

[0026] FIG. 5 is a schematic view of an air intake segment/an air intake tube according to some embodiments of the present disclosure.

[0027] FIG. 6 is a schematic view of a condensation segment/a condensation tube according to some embodiments of the present disclosure.

REFERENCE NUMERALS OF THE ACCOMPANYING DRAWINGS

- [0028] door body 1000; inner door 1100; first through hole 1110; outer door 1200;
- [0029] air discharge device 2000;
- [0030] air duct 2100; air intake segment/air intake tube 2110; air intake opening 2111; condensation segment/condensation tube 2120; air discharge segment/air discharge tube 2130; air discharge opening 2131; water return opening 2132;
- [0031] fan 2200;
- [0032] tub 3000; washing chamber 3100; bottom plate 3110; rear side plate 3120; top plate 3130.

DETAILED DESCRIPTION OF EMBODIMENTS

[0033] Technical solutions according to embodiments of the present disclosure will be described clearly and completely below in combination with accompanying drawings of the embodiments of the present disclosure. Obviously, the embodiments described below are only a part of the embodiments of the present disclosure, rather than all embodiments of the present disclosure. Based on the embodiments in the present disclosure, all other embodiments obtained by those skilled in the art without creative labor fall within the protection scope of the present disclosure.

[0034] It should be noted that all directional indications (such as up, down, left, right, front, rear, etc.) in the embodiments of the present disclosure are only used to explain relative positions between various components, movements of various components, or the like under a predetermined posture (as illustrated in the figures). When the predetermined posture changes, the directional indications also change accordingly.

[0035] In the present disclosure, unless otherwise clearly specified and limited, terms, such as “connect” “fix” or the like, should be understood in a broad sense. For example, it may be a fixed connection or a detachable connection or connection as one piece; mechanical connection or electrical connection; direct connection or indirect connection through

an intermediate; internal communication of two components or the interaction relationship between two components, unless otherwise clearly limited. For those of ordinary skill in the art, the specific meaning of the above-mentioned terms in the present disclosure can be understood according to specific circumstances.

[0036] In addition, descriptions, such as “first” “second” or the like, in the present disclosure are only used for descriptive purposes, and cannot be understood as indicating or implying relative importance or implicitly indicating the number of indicated technical features. Therefore, the features associated with “first” and “second” may explicitly or implicitly include at least one of the features. In addition, combinations can be performed on the technical solutions according to various embodiments of the present disclosure, but these combinations must be based on the fact that they can be realized by those skilled in the art. When a combination of the technical solutions is contradictory or unattainable, the combination of the technical solutions neither exists nor falls within the protection scope of the appended claims of the present disclosure.

[0037] A dishwasher is a device that can automatically clean dishes. Generally speaking, the dishwasher includes a base, a tub, and a door body. The tub is mounted on the base. The tub has a washing chamber. The door body is rotatably connected to the tub and/or the base for closing or opening the washing chamber. The washing chamber is provided with a dish rack that can be pulled out or pushed in. When the door body opens, the user can pull out the dish rack, load the dishes onto the dish rack, and then push the dish rack back into the washing chamber after loading. The user then closes the door body to seal the washing chamber, and the washing chamber can begin a washing process.

[0038] A spray arm is disposed in the washing chamber. The spray arm is in communication with an outlet of a washing pump of the base through a water path, and an inlet of the washing pump is in communication with a water cup. The water cup is located at a bottom of the tub and in communication with the washing chamber. Water in the washing chamber can flow to the water cup. When the dishwasher begins to operate, a water inlet valve of the dishwasher opens, allowing water to flow into the washing chamber. Once a water level in the washing chamber reaches a predetermined level, the water inlet valve closes, cutting off continued input of water. The dishwasher controls the washing pump to operate. The washing pump sucks water in the water cup and delivers the water to the spray arm. The spray arm has a nozzle. The water flow is ejected from the nozzle of the spray arm under an action of a pressure. The spray arm is urged to rotate under an action of a reverse driving force of the water flow ejected from the nozzle of the spray arm, in such a manner that the water flow is ejected onto the dishes to clean the dishes. The water sprayed by the spray arm flows back to the washing chamber and then flows to the water cup. The water cup is provided with a filter. The water flow continues to circulate under an action of the washing pump after being filtered by the filter until a current washing sequence is finished.

[0039] To improve a cleaning effect of the dishes, generally speaking, it is necessary to heat the water during the washing sequence. For example, a heating assembly is disposed in the washing pump. The washing pump may synchronously heat the water while sucking the water in the

washing chamber. That is, the water flow sprayed by the spray arm has a higher temperature, improving the cleaning effect on the dishes.

[0040] After a washing phase, the dishes need to be dried. Since the washing chamber of the tub is in a high-temperature and high-humidity environment after the washing stage is finished, in the related art, a dishwasher is provided with the air discharge device. The air discharge device directly extracts the air in the washing chamber and discharges the air to an external environment.

[0041] Since the air in the washing chamber in this case is at a high temperature and high humidity, the air discharged by the air discharge device has a high temperature, which poses a burn risk. In addition, when the high-temperature and high-humidity air is discharged to the external environment and encounters cooler temperatures, condensate water can form. For example, the condensate water can form at an air discharge outlet of the air discharge device. The condensate water can run down an exterior surface of the dishwasher to the floor. Or, when the expelled air encounters an obstacle, the condensate water can form on the obstacle, such as on the floor, which not only diminishes user experience, but also poses a safety risk. Therefore, the present disclosure provides a dishwasher, which is capable of reducing a probability of the condensate water forming outside the dishwasher, and reducing the air discharge temperature, and thus improving the user experience.

[0042] As illustrated in FIG. 1 to FIG. 4, in some embodiments of the present disclosure, the dishwasher includes a door body 1000 as a structure configured to close a tub 3000. The door body includes an outer door 1200 and an inner door 1100. The inner door 1100 and the outer door 1200 are connected to each other through some connection methods. For example, the outer door 1200 and the inner door 1100 are superimposed together, and a periphery of the inner door 1100 and a periphery of the outer door 1200 are connected to each other through screwing. Generally speaking, the inner door 1100 and the outer door 1200 are sheet metal parts, which are suitable for use in larger dishwashers and have high structural strength. However, the inner door 1100 and the outer door 1200 are not limited to the sheet metal parts, and may be made of other materials. For example, when the dishwasher is a small desktop machine, the inner door 1100 and the outer door 1200 may be made of plastic.

[0043] The inner door 1100 has a first through hole 1110. For example, the first through hole 1110 is a circular through hole. An air discharge device 2000 can be in communication with a washing chamber 3100 through the first through hole 1110, extracting air from the washing chamber 3100. It should be understood that the air discharge device 2000 in communication with the washing chamber 3100 through the first through hole 1110 can be that an air intake opening 2111 of the air discharge device 2000 passes through the first through hole 1110 to be in communication with the washing chamber 3100, or that the air intake opening 2111 of the air discharge device 2000 may be connected to the inner door 1100 to allow the air intake opening 2111 of the air discharge device 2000 to be in communication with the first through hole 1110, and the first through hole 1110 is in communication with the washing chamber 3100, realizing communication between the air intake opening 2111 of the air discharge device 2000 and the washing chamber 3100.

[0044] The air discharge device 2000 is a device configured to extract and discharge the air in the washing chamber

3100. The air discharge device 2000 is mounted between the inner door 1100 and the outer door 1200. That is, after the inner door 1100 is connected to the outer door 1200, a predetermined space between the inner door 1100 and the outer door 1200 is formed to place the air discharge device 2000. The air discharge device 2000 may be fixed only to the inner door 1100, may also be fixed only to the outer door 1200, or may also be fixed to both the inner door 1100 and the outer door 1200.

[0045] The air discharge device 2000 includes a fan 2200 which provides power for the air discharge device 2000 to extract and discharge the air in the washing chamber 3100, and an air duct 2100 which forms a path through which air flows. In an exemplary embodiment of the present disclosure, the air duct 2100 needs to be disposed to be in communication with the washing chamber 3100. As described above, the air duct 2100 may pass through the first through hole 1110 to achieve communication with the washing chamber 3100. For example, the air duct 2100 has the air intake opening 2111 in communication with the washing chamber 3100. The air duct 2100 has an air discharge opening 2131, through which the air duct 2100 is in communication with the external environment. The fan 2200 is disposed in the air duct 2100. When the fan 2200 is in operation, a negative pressure may be formed in the air duct 2100. Therefore, the air in the washing chamber 3100 can enter the air duct 2100 through the air intake opening 2111, then flow through the fan 2200 and continue to be discharged to the external environment through the air discharge opening 2131 of the air duct 2100. It should be understood that the fan 2200 may have various positions in the air duct 2100. For example, the fan 2200 is arranged close to the air intake opening 2111 and away from the air discharge opening 2131, that is, close to the first through hole 1110. Or, the fan 2200 may be arranged close to the air discharge opening 2131 and away from the air intake opening 2111, that is, away from the first through hole 1110. It is also possible that the fan 2200 is arranged between the air intake opening 2111 and the air discharge opening 2131 of the air duct 2100, as long as the air in the washing chamber 3100 can be extracted into the air duct 2100 and discharged from the air duct 2100.

[0046] The air duct 2100 has a water return opening 2132 configured to discharge the condensate water back into the washing chamber 3100. Since the high-temperature and high-humidity air in the washing chamber 3100 enters the air duct 2100, which is equivalent to the high-temperature and high-humidity air being cold, a predetermined amount of condensate water is formed in the air duct 2100. When the condensate water in the air duct 2100 is not treated in time, an efficiency of air extraction and discharge can be affected, and even the condensate water can flow out of the air discharge opening 2131. Therefore, with the water return opening 2132, when the condensate water is formed in the air duct 2100, the condensate water flows into the washing chamber 3100 through the water return opening 2132, and thus the condensate water flowing back into the washing chamber 3100 can be further discharged by a drain pump of the dishwasher. It should be understood that the tub 3000 of the dishwasher forms the washing chamber 3100. The water cup is disposed below the tub 3000 of the dishwasher, and in communication with the washing chamber 3100. That is, the condensate water flowing out of the water return opening 2132 may eventually flow to the water cup and then be

drained away through drainage. Or, the condensate water flowing out of the water return opening **2132** can directly flow to the water cup by some middle conduits. Since the water cup is in communication with the washing chamber **3100**, this situation is also regarded as the condensate water being discharged to the washing chamber **3100**.

[0047] In addition to discharging the condensate water into the washing chamber **3100**, the water return opening **2132** is also designed to allow external air to freely enter the air duct **2100**.

[0048] In particular, the water return opening **2132** is disposed upstream of the fan **2200**. It should be understood that, relative to the fan **2200**, the fan **2200** has an inlet and an outlet. Operation of the fan **2200** makes air enter from the inlet and be discharged from the outlet. A path through which the air flows to the inlet can be regarded as an upstream, and a path through which the air flows out from the outlet can be regarded as a downstream. Since the water return opening **2132** is disposed upstream of the fan **2200**, when the fan **2200** is in operation, the air in the air duct **2100** has a fast flow rate and a smaller pressure. Therefore, the external air (which has a lower temperature and humidity than the air in the washing chamber **3100**) can enter the air duct **2100** from the water return opening **2132** to mix with the high-temperature and high-humidity air from the washing chamber **3100** in the air duct **2100**, realizing mixing of air to reduce the air discharge temperature, preventing the air discharge temperature from being too high and causing harm to the user, especially young children.

[0049] In addition, relative to the high-temperature and high-humidity air in the washing chamber **3100**, the external air has a relatively lower temperature and lower humidity. In this way, when the two air streams are mixed in the air duct **2100**, a predetermined amount of condensate water may occur, which can also be discharged from the air duct **2100** through the water return opening **2132** to avoid flowing out at the air discharge opening **2131**. Since the high-temperature and high-humidity air from the washing chamber **3100** can reduce moisture in the air after being mixed with the external air, the air discharged through the air discharge opening **2131** is less likely to form the condensate water in the external environment, reducing a probability of forming the condensate water in the external environment, improving the user experience.

[0050] The fan **2200** extracts the air in the washing chamber **3100** and discharges the air to the external environment through the air discharge opening **2131**. The air discharge opening **2131** may be mounted at various positions, such as drilling a hole on a front surface of the outer door **1200** for the air discharge opening **2131** to discharge the air, or drilling a hole on a side surface of the outer door **1200** for the air discharge opening **2131** to discharge the air. However, these discharging air methods involve drilling a hole in the user's line of sight, which is likely to compromise overall integrity of the door body **1000**. For this reason, in another exemplary embodiment of the present disclosure, the air discharge opening **2131** of the air discharge device **2000** is disposed below the door body **1000**, for example, between the door body **1000** and a toe kick board. That is, the dishwasher has the door body **1000** and the toe kick board located below the door body **1000** at a front side of the dishwasher. The toe kick board is disposed to avoid being too hollowed out under the door body **1000**, and the dishwasher has a more complete front side. Since the door body

1000 needs to be set as rotatable, there is a predetermined gap between the door body **1000** and the toe kick board. The air discharge opening **2131** may be in the gap between the door body **1000** and the toe kick board and opened towards a front of the dishwasher (a side close to the user is the front side when the user uses the dishwasher, that is, an orientation where the door body **1000** is located is the front side, and the door body **1000** rotates from top to bottom for opening the washing chamber **3100**). In this way, the air discharge opening **2131** is hidden in the user's line of sight without affecting air discharge.

[0051] As can be understood, the fan **2200** is a device capable of providing power for extracting and discharging the air. For example, the fan **2200** includes a motor and an impeller drivingly connected to the motor. The impeller is disposed in the air duct **2100** to drive the air to flow through the air duct **2100**. In another exemplary embodiment of the present disclosure, the impeller adopts a structure of a centrifugal impeller to allow the air duct **2100** to be made flatter, reducing a space between the outer door **1200** and the inner door **1100** occupied by the air discharge device **2000**.

[0052] As can be seen from the above, by providing the air discharge device **2000** between the outer door **1200** and the inner door **1100**, the air in the washing chamber **3100** is effectively extracted and discharged, and thus the drying of the dishes is realized. The air discharge device **2000** has the water return opening **2132**. The water return opening **2132** is configured to discharge the condensate water, and when the fan **2200** is in operation, the external air can enter the air duct **2100** through the water return opening **2132** and mix with the high-temperature and high-humidity air extracted into the washing chamber **3100** in the air duct **2100**. In this way, the air discharge temperature of the air discharge device **2000** and a risk can be reduced, and moisture in the air when the air is mixed can be further removed. The formed condensate water is discharged back to the washing chamber **3100** through the water return opening **2132**, so as to reduce the probability of generation of the condensate water in the external environment. By disposing the water return opening **2132**, both recirculation of condensate water and mixing of air to lower the air discharge temperature are realized.

[0053] As illustrated in FIG. 3 and FIG. 4, in some embodiments of the present disclosure, to facilitate outflow of the condensate water from the water return opening **2132**, the water return opening **2132** is open downwardly in this embodiment. Where the "open downwardly" can be directly below or obliquely below, and the "open" means forming an opening to be in communication with the air duct **2100**. In this way, when the condensate water is formed in the air duct **2100**, the condensate water can flow out from the water return opening **2132** under the action of gravity, without a need for additional guide structures or extraction devices for extracting. This design prevents accumulation of the condensate water and makes an overall structure of the air discharge device **2000** simpler and more reliable, thereby reducing cost of the product.

[0054] For example, when the air from the washing chamber **3100** enters the air duct **2100**, the high-temperature and high-humidity air encounters a wall of the air duct **2100** and is cooled to form the condensate water on the wall of the air duct **2100**, and the condensate water flows along the wall of the air duct **2100** to the water return opening **2132** under the action of gravity and is discharged through the water return

opening 2132. However, the air entering from the water return opening 2132 and the high-temperature and high-humidity air from the washing chamber 3100 collide and create turbulence in the air duct 2100, which causes the condensate water to form on corresponding walls of the air duct 2100 and flow out through the water return opening 2132 under the action of gravity.

[0055] It should be understood that when the condensate water is discharged from the water return opening 2132, the air in the air duct 2100 is still in a flowing state in this case. Therefore, the external air can enter the air duct 2100 from the water return opening 2132 under the action of a pressure. That is, discharging of the condensate water and entry of the external air are performed simultaneously. The discharging of the condensate water does not affect the entry of the external air into the air duct 2100 through the water return opening 2132, and the entry of the external air into the air duct 2100 through the water return opening 2132 does not affect the discharging of the condensate water, in such a manner that the extraction efficiency of the air in the washing chamber 3100 can be improved, and thus a drying efficiency of the dishes can be improved.

[0056] In some embodiments of the present disclosure, as illustrated in FIG. 1 and FIG. 2, the dishwasher includes the tub 3000 forming the washing chamber 3100. For example, the dishwasher has a top plate 3130, a rear side plate 3120, a left side plate, a right side plate, and a bottom plate 3110. The left side plate is disposed at a side of the rear side plate 3120. The right side plate is disposed at another side of the rear side plate 3120. The top plate 3130 is disposed at a top of the rear side plate 3120, the left side plate, and the right side plate. The bottom plate 3110 is disposed at a bottom of the rear side plate 3120, the left side plate, and the right side plate. In this way, a washing chamber 3100 that is open towards the front is formed. The door body 1000 is rotatably disposed in front of the tub 3000 for opening or closing the washing chamber 3100. In this way, the water return opening 2132 is disposed above the bottom plate 3110 and spaced apart from the bottom plate 3110, in such a manner that the condensate water flowing out of the water return opening 2132 can be received by the bottom plate 3110 from top to bottom to flow into the washing chamber 3100, which is simple and convenient and effectively realizes backflow of the condensate water.

[0057] In an exemplary embodiment of the present disclosure, the water return opening 2132 is disposed above the bottom plate 3110. The “above” here may be directly above or obliquely above, as long as it is ensured that the condensate water flows to the bottom plate 3110 under the action of gravity and is received by the bottom plate 3110. For example, when the water return opening 2132 is located directly above the bottom plate 3110, that is, in a top-down projection of the dishwasher, a projection of the water return opening 2132 is located in a projection of the bottom plate 3110, a front end of the bottom plate 3110 may extend forwardly to extend between the inner door 1100 and the outer door 1200. In this way, the condensate water from the water return opening 2132 can flow back into the bottom plate 3110 in a shorter and faster path and be received by the bottom plate 3110, improving a discharge efficiency of the condensate water and avoiding generation of the condensate water in the door body 1000 due to the external air exchanging heat with the condensate water during the condensate water flowing back to the washing chamber 3100.

[0058] When the water return opening 2132 is not located directly above the bottom plate 3110 but is located obliquely above the bottom plate 3110, backflow of the condensate water to the washing chamber 3100 can also be realized, which is slightly inconvenient as compared to a situation where the water return opening 2132 is located directly above the bottom plate 3110. For example, in the top-down projection of the dishwasher, the projection of the water return opening 2132 is not located in the projection of the bottom plate 3110, but in front of the projection of the bottom plate 3110, in such a manner that the water return opening 2132 needs to guide the condensate water onto the bottom plate 3110 through some guide structures to be received by the bottom plate 3110. This more complicated guide structure may increase design difficulty of the air discharge device 2000, and may occupy too much space of the outer door 1200 and the inner door 1100, and even make a thickness of the door body 1000 increase.

[0059] In another exemplary embodiment of the present disclosure, as illustrated in FIG. 1 and FIG. 2, in some embodiments, the water return opening 2132 is designed to be spaced apart from the bottom plate 3110 without any direct contact, which further simplifies the structure, and avoids increasing complexity of the air discharge device 2000 by mounting of intermediate pieces, enabling the condensate water to return to the washing chamber 3100 at a fastest speed, and thus avoiding condensation of air in the door body 1000 during the backflow of the condensate water.

[0060] For example, when the water return opening 2132 is located directly above the bottom plate 3110, the water return opening 2132 may be designed to be close to the top of the bottom plate 3110 and in non-contact with the bottom plate 3110. In this way, the condensate water flowing out of the water return opening 2132 is separated from the water return opening 2132 under the action of gravity and drips onto the bottom plate 3110. The condensate water is only subjected to resistance of air but not by other resistance in a process of dripping onto the bottom plate 3110, improving the backflow efficiency of the condensate water and avoiding condensation of air in the door body 1000. In addition, after a drying stage of the dishwasher is completed, there is no residual condensation water in a condensation water dripping path from the water return opening 2132 onto the bottom plate 3110, and there is no possibility that condensation water flows to other parts of the door body 1000 after the door body 1000 of the dishwasher is rotated and opened.

[0061] For another example, when the water return opening 2132 is located obliquely above the bottom plate 3110, a guide groove needs to be designed between the water return opening 2132 and the bottom plate 3110. The condensate water dripping from the water return opening 2132 drips to the guide groove and is guided to the bottom plate 3110 through the guide groove. Compared to a scenario where the condensate water drips directly onto the bottom plate 3110, this method introduces an additional guide groove, which creates resistance to flow of the condensate water, reducing the backflow efficiency of the condensate water. In addition, this method increases complexity of the outer door 1200 and the inner door 1100 to some extent.

[0062] As described above, to reduce a temperature and humidity of the high-temperature and high-humidity air from the washing chamber 3100 in the air duct 2100, the entry of the external air is realized by disposing the water

return opening 2132. It should be understood that the external air has a lower temperature and lower humidity than the high-temperature and high-humidity air of the washing chamber 3100. To facilitate the entry of the external air into the air duct 2100 through the water return opening 2132, as illustrated in FIG. 1 and FIG. 2, in some embodiments, the water return opening 2132 is in communication with the space between the outer door 1200 and the inner door 1100. That is, the water return opening 2132 is disposed between the inner door 1100 and the outer door 1200 and in direction communication with the space between the inner door 1100 and the outer door 1200. Further, generally speaking, the inner door 1100 and the outer door 1200 of the dishwasher are not sealed together. That is, the space enclosed by the inner door 1100 and the outer door 1200 is in communication with an external space outside the dishwasher. Therefore, when the air enters the air duct 2100 through the water return opening 2132, air from the external space outside the dishwasher may also enter the space between the inner door 1100 and the outer door 1200.

[0063] The water return opening 2132 is in communication with the space between the outer door 1200 and the inner door 1100, in such a manner that the entire air discharge device 2000 can be hidden between the inner door 1100 and the outer door 1200. An existing non-sealed connection structure between the outer door 1200 and the inner door 1100 can be fully utilized to realize entry of cold air. An opening on the outer door 1200 can be avoided to make the cold air enter the air duct 2100, resulting in a simpler structure and aesthetic appearance of the door body 1000.

[0064] In another exemplary embodiment of the present disclosure, as illustrated in FIG. 3 and FIG. 4, in some embodiments, the water return opening 2132 is disposed below the inlet of the fan 2200, which is more conducive to the discharging of the condensate water and prevents the condensate water from flowing out by the fan 2200.

[0065] For example, the condensate water is formed on the wall of the air duct 2100. The condensate water may flow towards the water return opening 2132 under the action of gravity, and then flow out through the water return opening 2132. Since the inlet of the fan 2200 has a higher position than the water return opening 2132, the condensate water is more likely to cross the inlet of the fan 2200 and flow through the water return opening 2132 under the action of gravity, and less likely to form convergence at the inlet of the fan 2200, avoiding entering the fan 2200.

[0066] As illustrated in FIG. 3 and FIG. 4, in some embodiments, an open direction of the water return opening 2232 spatially intersects with an air intake direction of the fan 2200, to prevent the condensate water from entering the fan 2200 to the greatest extent.

[0067] For example, when the condensate water is formed in the air duct 2100 and flows to the water return opening 2132 in a direction of gravity, a flow direction of the air in the air duct 2100 is roughly the same as a flow direction of the condensate water under an action of the fan 2200. That is, flow of the air in the air duct 2100 causes the condensate water to flow faster. Since the air intake direction of the fan 2200 intersects with the open direction of the water return opening 2132, when the condensate water flows to a position opposite to the fan 2200, it is equivalent to a sudden change in the intake air of the fan 2200. In this way, the condensate water can continue to flow to the water return opening 2132

under an action of inertia, reducing a likelihood of the condensate water being carried into the fan 2200 by an airflow. This design prevents the condensate water from being discharged through the air discharge opening 2131. For example, as illustrated in FIG. 3 and FIG. 4, the open direction of the water return opening 2132 is from top to bottom, and the air intake direction of the fan 2200 is perpendicular to a plane of paper/screen.

[0068] It should be understood that the fan 2200 includes the impeller, for example, a centrifugal impeller. The air intake direction of the fan 2200 is an extension direction of a rotational axis of the centrifugal impeller.

[0069] In another exemplary embodiment of the present disclosure, a main purpose of the fan 2200 is to extract the air in the washing chamber 3100. To effectively introduce the external air into the air duct 2100 through the water return opening 2132, the water return opening 2132 is arranged adjacent to the fan 2200, in such a manner that when the fan 2200 operates, a path from the water return opening 2132 to the inlet of the fan 2200 is shortened, enabling the external air to be more effectively introduced into the air duct 2100. Here, adjacent arrangement can be understood as being close to each other, for example, 3 cm, 5 cm, 8 cm, etc. apart.

[0070] As mentioned above, the external air fed into the air duct 2100 through the water return opening 2132 has the lower temperature and the lower humidity, which can neutralize the air having a higher temperature and higher humidity from the washing chamber 3100 in the air duct 2100. However, the air from the washing chamber 3100 has a large amount under the action of the fan 2200. If the amount of the external air entering from the water return opening 2132 is to be increased, a size of the water return opening 2132 needs to be increased. In this case, the extract effect on the washing chamber 3100 may be reduced. Therefore, before the air from the washing chamber 3100 meets the external air entering through the water return opening 2132, the air needs to be condensed. This pre-reduction of the temperature and humidity of the air from the washing chamber 3100 ensures that, when the air mixes with the external air from the water return opening 2132, the temperature and the humidity of the air can be minimized as much as possible. As illustrated in FIG. 3, in some embodiments, the air duct 2100 includes a condensation segment 2120. The so-called condensation segment 2120 can make the air from the washing chamber 3100 be condensed to a certain extent, reducing moisture and a temperature in the air. That is, the condensation segment 2120 needs to be provided with a condensation structure. For example, a structure of the condensation segment 2120 is set as a concave-convex shape to increase a heat exchange area, and thus the air in the air duct 2100 is effectively condensed.

[0071] The condensation segment 2120 extends in an up-down direction, in such a manner that the condensate water generated in the condensation segment 2120 flows downwardly under the action of gravity. When the water return opening 2132 is disposed below the condensation segment 2120, the condensate water can more easily flow to the water return opening 2132 and be discharged. In another exemplary embodiment of the present disclosure, an open direction of the water return opening 2132 is the same as an extension direction of the condensation segment 2120. In this way, the condensate water can be directly discharged through the water return opening 2132, preventing the

condensate water from forming a circuitous path and accelerating the discharge efficiency of the condensate water. For example, as illustrated in FIG. 3 and FIG. 6, the condensation segment **2120** has a long strip structure extending in the up-down direction. The water return opening **2132** is disposed below the condensation segment **2120** and opens downwardly, in such a manner that the condensate water can flow out of the water return opening **2132** from top to bottom. In particular, when the condensate water generated in the condensation segment **2120** is separated from the wall of the condensation segment **2120**, the condensate water can be directly discharged through the water return opening **2132** under the action of the gravity and the airflow, effectively accelerating a drainage efficiency of the condensate water.

[0072] As illustrated in FIG. 3, FIG. 5 and FIG. 6, in some embodiments, the air duct **2100** includes an air intake segment **2110**, a condensation segment **2120**, and an air discharge segment **2130** that are sequentially in communication with each other. The air intake segment **2110** has the air intake opening **2111**. The air discharge segment **2130** has the air discharge opening **2131**. The air intake segment **2110** is in communication with the washing chamber **3100** through the air intake opening **2111**. The fan **2200** is disposed in the air discharge segment **2130**. When the fan **2200** is in operation, the air from the washing chamber **3100** enters the air intake segment **2110**, then flows through the condensation segment **2120** and the air discharge segment **2130**, and is ultimately discharged. The air intake segment **2110** is constructed as an inverted U-shape. In this way, when the dishwasher is in the washing state, even if the water flow sprayed by the spray arm of the dishwasher enters the air intake segment **2110** through the air intake opening **2111**, the water entering through the air intake opening **2111** will flow back into the washing chamber **3100** instead of entering the condensation segment **2120**, because the air intake segment **2110** is an inverted U-shape, and a part of the air intake segment **2110** having the air intake opening **2111** extends from bottom to top.

[0073] The water return opening **2132** may be disposed in the air discharge segment **2130** or the condensation segment **2120**. Since the main function of the condensation segment **2120** is condensation, the water return opening **2132** disposed in the air discharge segment **2130** can avoid adverse effects on air condensation.

[0074] In an exemplary embodiment of the present disclosure, the air duct **2100** includes an air intake tube **2110** forming the air intake segment **2110**, a condensate tube **2120** forming the condensation segment **2120**, and an air discharge tube **2130** forming the air discharge segment **2130**. The air intake tube **2110**, the condensate tube **2120**, and the air discharge tube **2130** are manufactured separately and then connected together. The air intake tube **2110** and the condensate tube **2120**, and the condensate tube **2120** and the air discharge tube **2130** can be connected to each other by a variety of connection methods, such as insertion, snap, welding, or screwing, which makes manufacturing of the entire air duct **2100** more convenient. For the air intake tube **2110**, the condensate tube **2120**, and the air discharge tube **2130**, each can be formed by connecting two pieces, for example, by welding the two pieces. In this way, a hollow structure for the air flow can be created.

[0075] Through condensation of the condensation segment **2120** and introducing the external air through the water

return opening **2132**, the moisture of the air discharged from the air duct **2100** is greatly reduced, and a probability of forming condensate water in the air discharge opening **2131** and in the external environment is greatly reduced. However, the final discharged air still contains moisture and cannot achieve complete drying. To further reduce formation of the condensate water, as illustrated in FIG. 2, in some embodiments, the fan **2200** is located below $\frac{1}{2}$ or $\frac{1}{3}$ of a height of the inner door **1100**. That is, a distance from the fan **2200** to a bottom edge of the inner door **1100** (excluding hinges on two sides of the inner door **1100**) is smaller than or equal to $\frac{1}{2}$ or $\frac{1}{3}$ of the height of the inner door **1100**, resulting in a shorter path from the fan **2200** to the air discharge opening **2131**. The shorter path reduces a probability of air condensing in this path, allowing the air to be discharged more quickly into the external environment to further avoid the probability of formation of the condensate water at the air discharge opening **2131**. In addition, the air discharged into the external environment, having significantly reduced moisture, is less likely to form the condensate water when the air mixes thoroughly with the external air.

[0076] Although some embodiments of the present disclosure are described above, the scope of the present disclosure is not limited to the embodiments. Under the concept of the present disclosure, any equivalent structure transformation made using the contents of the specification and the accompanying drawings of the present disclosure, or any direct or indirect application of the contents of the specification and the accompanying drawings of the present disclosure in other related fields, shall equally fall within the scope of the present disclosure.

What is claimed is:

1. A dishwasher, comprising:

an outer door;

an inner door connected to the outer door and having a first through hole; and

an air discharge device disposed between the outer door and the inner door, the air discharge device comprising an air duct and a fan, the air duct being in communication with a washing chamber of the dishwasher through the first through hole, the fan being adapted to draw air from the washing chamber into the air duct and discharge the air out of the air duct, the air duct having a water return opening upstream of the fan, and the water return opening being configured to allow external air to enter the air duct from the water return opening during an operation of the fan and to discharge condensate water into the washing chamber.

2. The dishwasher according to claim 1, further comprising a tub, the tub comprising a bottom plate enclosing the washing chamber, and the water return opening being located above the bottom plate and spaced apart from the bottom plate.

3. The dishwasher according to claim 2, wherein the water return opening is located directly above the bottom plate.

4. The dishwasher according to claim 3, wherein the condensate water drips from the water return opening onto the bottom plate.

5. The dishwasher according to claim 1, wherein the water return opening is open downwardly.

6. The dishwasher according to claim 1, wherein the water return opening is in communication with a space between the inner door and the outer door.

7. The dishwasher according to claim 1, wherein:
the water return opening is located below an inlet of the fan; and/or
an open direction of the water return opening spatially intersects with an air intake direction of the fan.
8. The dishwasher according to claim 1, wherein the water return opening is arranged adjacent to the fan.
9. The dishwasher according to claim 1, wherein the air duct includes a condensation segment extending in an up-down direction, the water return opening being located below the condensation segment.
10. The dishwasher according to claim 9, wherein an open direction of the water return opening is the same as an extension direction of the condensation segment.
11. The dishwasher according to claim 1, wherein the air duct comprises an air intake segment, a condensation segment, and an air discharge segment that are sequentially in communication with each other, the air intake segment being

constructed as an inverted U-shape and in communication with the washing chamber, the fan being disposed in the air discharge segment, and the water return opening being disposed in one of the air discharge segment and the condensation segment.

12. The dishwasher according to claim 11, wherein the air duct comprises an air intake tube forming the air intake segment, a condensate tube forming the condensation segment, and an air discharge tube forming the air discharge segment, and the fan and the water return opening being located in the air discharge segment.

13. The dishwasher according to claim 11, wherein:
the fan is located below $\frac{1}{2}$ or $\frac{1}{3}$ of a height of the inner door; and
the air discharge segment has an air discharge opening located below the inner door and the outer door.

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